Problem on Electrical Energy and Power / Solution

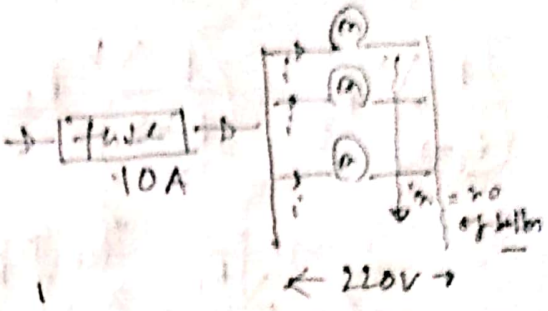
(1) find across each bulb 220 V

∵ power of each bulb is 100W so

current flow each bulb $i = \frac{P}{V} = \frac{100}{220}$ A

max^m main current = 10A

Let max^m no of bulbs in parallel be n



$\therefore 10A = n i$

$10 = n \times \frac{100}{220}$

$\boxed{n = 22}$

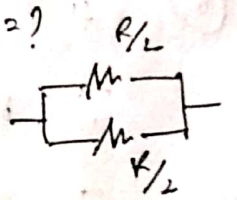
(2) $P = \frac{V^2}{R}$ ∵ V = const (220V)

$\therefore \frac{P_2}{P_1} = \frac{R_1}{R_2}$

$\frac{P_2}{100} = \frac{R}{R/4}$

$\frac{P_2}{100} = 4 \Rightarrow \boxed{P_2 = 400W}$

$P_2 = ?$



$R_2 = \frac{R/2 \times R/2}{R/2 + R/2} = R/4$

(3) $P = \frac{V^2}{R}$ ∵ R = constant

$\therefore \frac{V^2}{P}$ = const

$\therefore \frac{V_2^2}{P_2} = \frac{V_1^2}{P_1} \Rightarrow P_2 = \left(\frac{V_2}{V_1}\right)^2 P_1$

$= \left(\frac{110}{220}\right)^2 \times 1000$

$= \frac{1000}{4}$

$\boxed{P_2 = 250W}$

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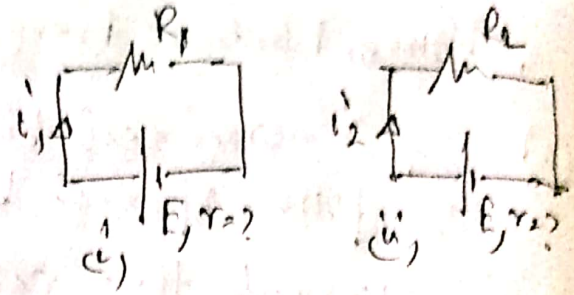
(4)



$$P_1 = P_2$$

$$i_1^2 R_1 = i_2^2 R_2$$

$$\left(\frac{E}{R_1+r}\right)^2 R_1 = \left(\frac{E}{R_2+r}\right)^2 R_2 \Rightarrow \frac{R_1}{(R_1+r)^2} = \frac{R_2}{(R_2+r)^2}$$

$$\Rightarrow \frac{\sqrt{R_1}}{R_1+r} = \frac{\sqrt{R_2}}{R_2+r}$$

$$r\sqrt{R_1} + \sqrt{R_1} \times R_2 = r\sqrt{R_2} + \sqrt{R_2} \times R_1$$

$$r\left(\sqrt{R_1} - \sqrt{R_2}\right) = \sqrt{R_1 R_2}\left(\sqrt{R_1} - \sqrt{R_2}\right)$$

$$\boxed{r = \sqrt{R_1 R_2}}$$

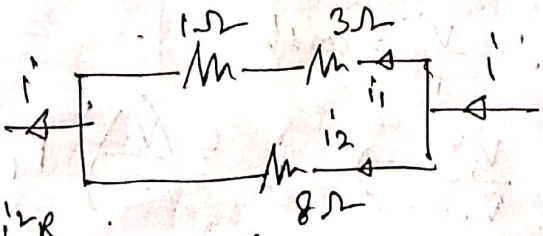
(5) $H = P_t t \Rightarrow H = \frac{V^2 t}{R}$

$\because$ H, V are const $\therefore \frac{t}{R}$ = constant

$$\therefore \frac{t_{series}}{t_{parallel}} = \frac{R_{series}}{R_{parallel}}$$

$$\frac{t_s}{6} = \frac{3R}{R/3} \Rightarrow t_s = 6 \times 9 = 54\ min$$

(6)



$\because$ power dissipated in 8Ω is 2W so

$P = i^2 R$

$i_2^2 \times 8 = 2$

$i_2 = \frac{1}{2} A$ → Now, (1+3) ie 4Ω & 8Ω are in parallel

So $i_1 \times 4 = i_2 \times 8$

$i_1 = 2 i_2$

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$\therefore \quad i_1 = 2 \times \frac{1}{2}$

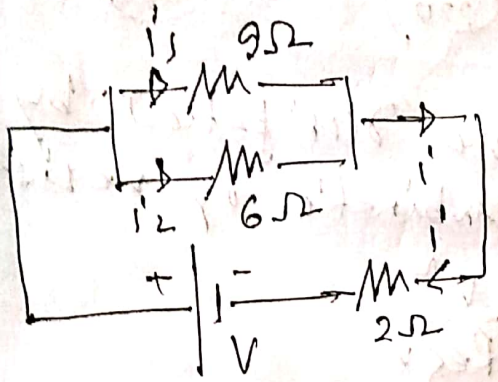
$= \underline{1A.}$

So power dissipated in 3Ω is,

$P_3 = i_1^2 \times R$

$= (1)^2 \times 3 = \underline{3W}$

(7)



Power in 9Ω

$= i_1^2 \times 9 = 36$

$i_1^2 = 4 \quad \underline{i_1 = 2A}$

voltage acrs 9Ω = voltage acrs 6Ω

$i_1 \times 9 = i_2 \times 6$

$2 \times 9 = i_2 \times 6 \Rightarrow \underline{i_2 = 3A}$

$\therefore \quad i = i_1 + i_2 = 3+2 = \underline{5A}.$

So p.d acrs 2Ω is $V = iR$

$= 5 \times 2$

$= \underline{10V}$

(8)

$P = \frac{V^2}{R}, \quad R = \text{constant}$

$\ln P = 2 \ln V - \ln R$

taking differential both sides,

$\frac{dP}{P} = \frac{2dV}{V} \Rightarrow \frac{dP}{P} \times 100 = 2\left(\frac{dV}{V} \times 100\right)$

$\therefore \frac{dP}{P} \times 100 = 2 \times 2.5$

$= \underline{5\%}$

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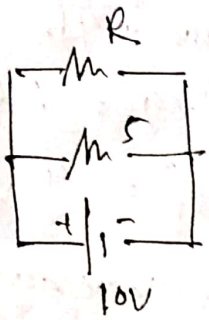
(9) $P = \frac{V^2}{R}$, here $P = 60\,W$ is constant

$$\therefore \frac{V_2^2}{R_2} = \frac{V_1^2}{R_1}$$

$$R_2 = R_1 \times \left(\frac{V_2}{V_1}\right)^2$$

$$= R \times \left(\frac{110}{220}\right)^2 = \left(\frac{R}{4}\right)\Omega$$

(10)



Net Power dissipated by any combination of resistors $= \frac{V^2}{R_{eqv}}$

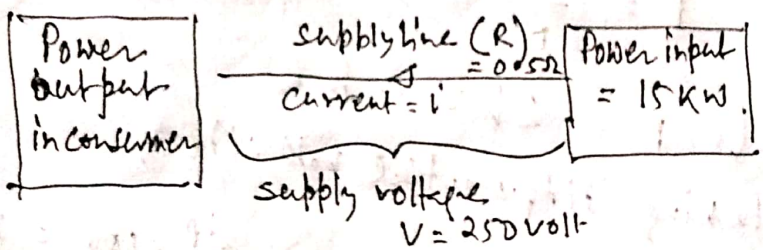
$$\therefore P_{net} = \frac{V^2}{\left(\frac{5R}{5+R}\right)} = 30$$

$$\Rightarrow (10)^2 \times \left(\frac{5+R}{5R}\right) = 30$$

$$\frac{5+R}{R} = \frac{3}{2}$$

$$10 + 2R = 3R \Rightarrow \boxed{R = 10\Omega}$$

(11)



Power is lost in supply line as heat

$$\therefore P_{lost} = i^2 R$$

$$= (60)^2 \times 0.5$$

$$= 36 \times 100 \times \frac{5}{10}$$

$$= 1800\,W$$

$$= 1.8\,KW$$

$$P_{in} = V_{in}\, i$$

$$i = \frac{P_{in}}{V_{in}} = \frac{15 \times 10^3}{250} = 60\,A$$

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So Poutput

Power output (at consumer) = Pinput − Ploss in supply line as heat

= 15 − 1.8

= 13.2 KW

∴ efficiency of supply

$= \frac{P_{out}}{P_{in}} \times 100 = \frac{13.2}{15} \times 100$

$= \frac{132}{3} \times \frac{20}{10}$

$= 88\%$

Note [extra knowledge]

Power lost in supply line as heat $= i^2 R$

$$\boxed{P_{loss\ in\ cable} = \frac{(P_{in})^2 R}{V_{supply}}}$$

$\begin{cases} P_{input} = V_{supply} \times i \\ i = \frac{P_{in}}{V_{supply}} \end{cases}$

∵ R of cable is constant. P_{in} = const

∴ Ploss will decrease if Vsupply is increased.

For this reason electric power from power plant is delivered to consumer at very high voltage (and consequently at very low current ∵ $V \times i = P_{in}$ = constant). So that power lost as heat in supply cable is minimum.

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12 no. of kWhs or no of B.O.T. units consumed

$$= \frac{\text{watt} \times \text{hr}}{1000}$$

April = 30 days

$$= \frac{\left[\overbrace{(6\times40 + 2\times100)}^{\text{bulbs}}\times5 + \underbrace{(4\times40)}_{\text{fans}}\times15 + \underbrace{1000\times2}_{\text{heater}}\right]\times30}{1000}$$

$$= 66 + 72 + 60$$

$$= \underline{198}$$

$$\therefore \text{Bill} = 198 \times \frac{3}{2} = 99\times3$$

$$= \underline{297}$$

$$P = V\times i$$

$$i = \frac{P}{V} = \frac{\overbrace{(6\times40)}^{\text{bulb}} + \overbrace{2\times100}^{\text{bulb}} + \overbrace{(4\times40)}^{\text{fan}} + \overbrace{1000}^{\text{heater}}}{200}$$

$$= \frac{1600}{200} = \underline{8A}$$

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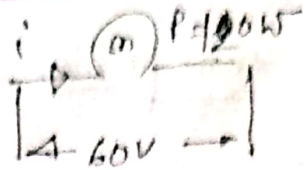
In 1st case

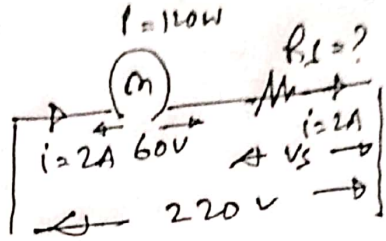
(13) When the bulb glows brightest i.e dissipates its full power $P = 60W$, current thro'



(i)

it $i = \frac{P}{V} = \frac{120}{60} = 2A$

In 2nd case,



Let resistance in series be $R_s\ \Omega$. ∵ the bulb still consumes 120W (glows fullest) so voltage across it is still 60V, So current thro' it & series resistance is still 2A.

Here voltage across series resistance $= 220 - 60 = 160$ volt

$\therefore R_s = \frac{V_s}{i} = \frac{160}{2} = 80\ \Omega$

(14) Type →

$W_{el} = \text{Heat}$

VIt or I^2Rt or, $\frac{V^2}{R}t$ $= m S_w \Delta\theta$

{S_w in SI $= 4200\ J/kg/°C$}

Here: $\frac{V^2}{R}t = m S_w \Delta\theta$

$\frac{220 \times 220}{12} \times t = 40 \times 4200 \times (80-15)$

$t = \frac{40 \times 4200 \times 65 \times 12}{220 \times 220}$

$= 2730$ sec (approx)

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